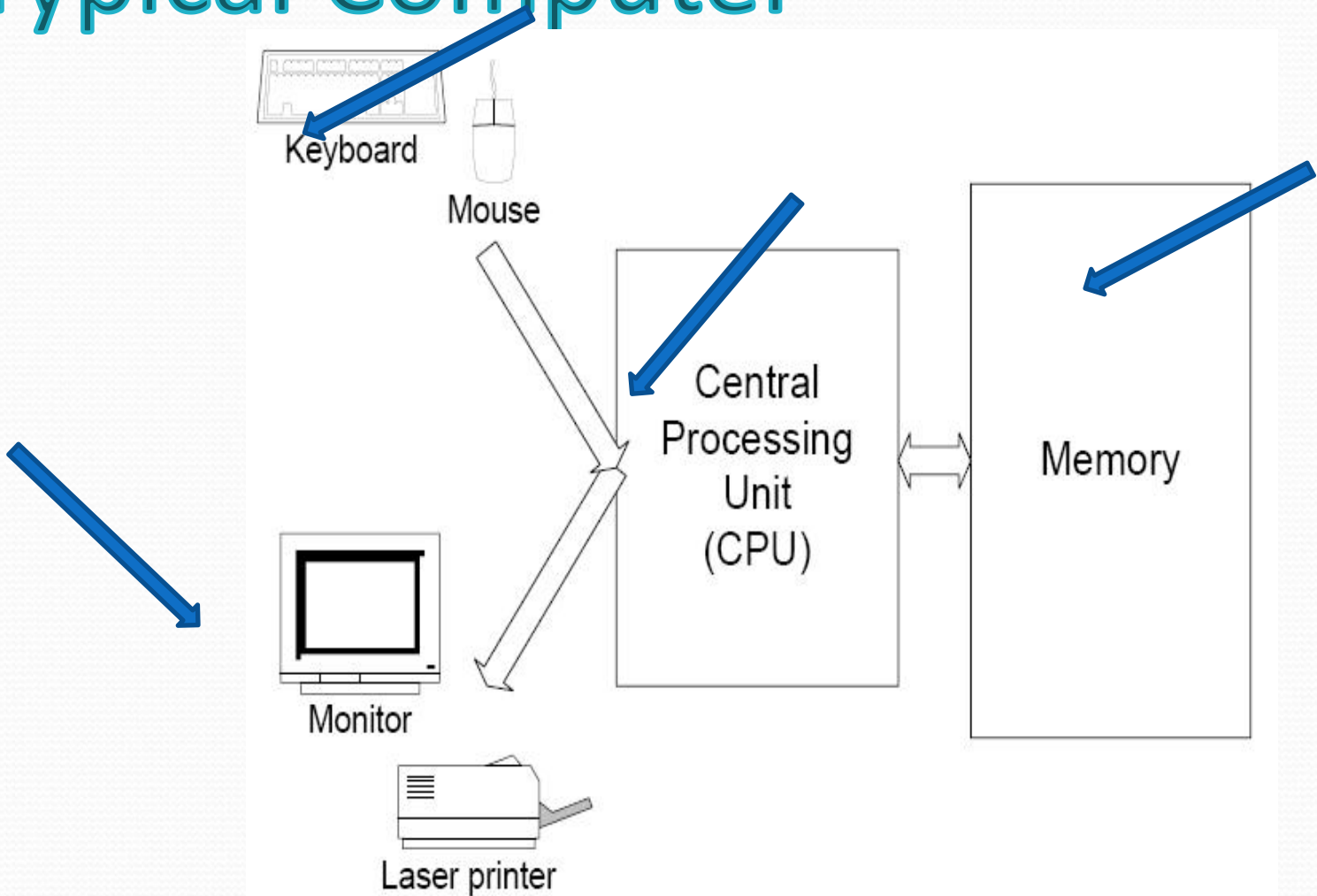


Computer Basics

- Computer is an electronic machine that stores, retrieves and processes data.
- Performs computations and makes logical decisions
- It cannot think or reason
- It can only carry out instructions given to it
- Millions / billions times faster than human beings

A Typical Computer



Work flow of Computer



- Input data from disk, keyboard or other storage media
- Process data
- Output results to display screen, disk or other media

Programming Languages

- **Computer programs**
 - Sets of instructions which directs a computer to perform certain function.

Types:

- Machine language
- Assembly language
- High-level languages

Machine language

- Binary
- Machine dependent
- Stored in the computer when the program is running
- A CPU can only execute machine language instructions - computer's language
- Example:
01110110001010010010

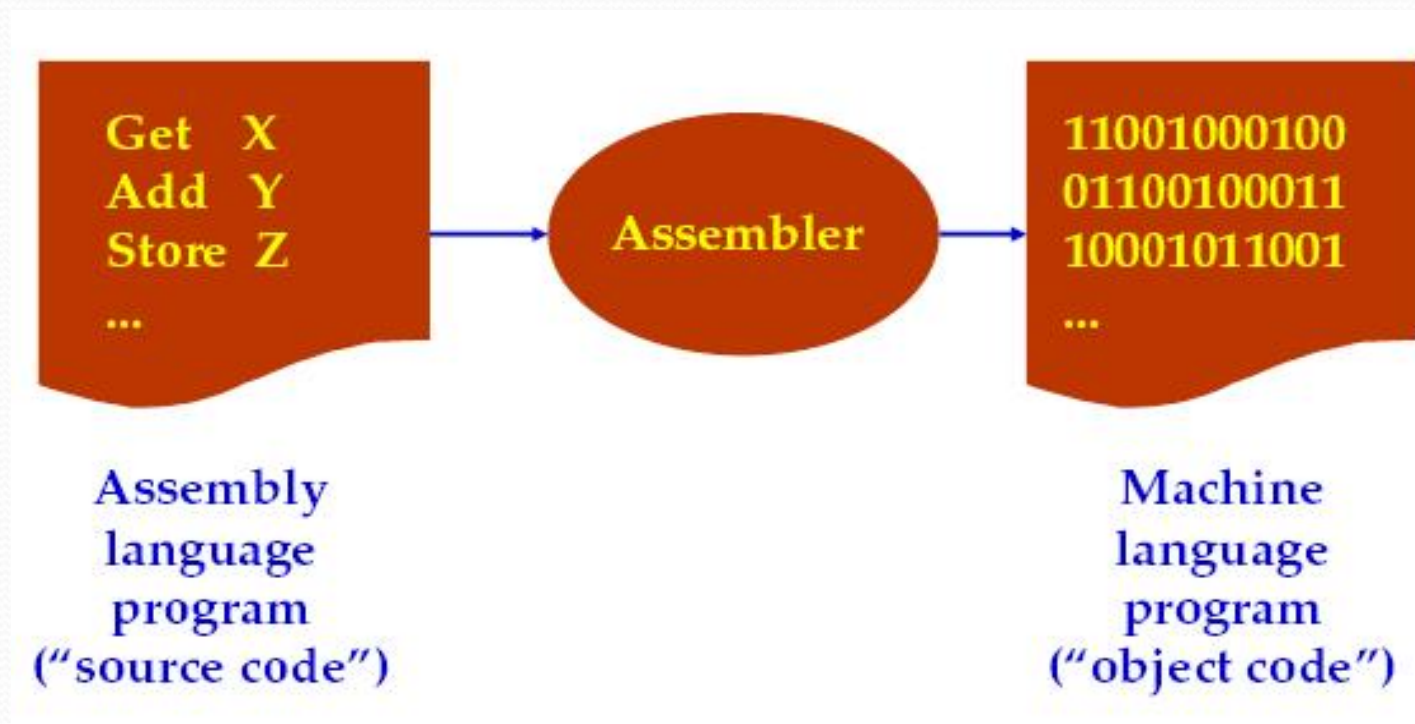
Assembly language

- Symbolic addressing, i.e.,
 - Use Symbolic names to represent operations, registers and memory locations.
- One-to-one correspondence with machine language
- A program written in assembly language must be converted to machine language before the CPU can execute it.
- A program called the assembler translates each assembly language statement into single machine language instruction.

Example:

- MOV AX,A
- ADD AX,4
- STORE Z

Automatically translating assembly language to machine language



High-level languages

- Closer to how people think about their problems
- No one-to-one correspondence to machine language
- General purpose
- Example:

$$Z = A + 4$$

High Level Language

- A "high-level language" is a programming language designed to be easily understood by humans, using commands and structures closer to natural language, making it easier to write code compared to low-level languages which are closer to machine code and require more technical knowledge; examples of high-level languages include Python, Java, C++, and JavaScript.



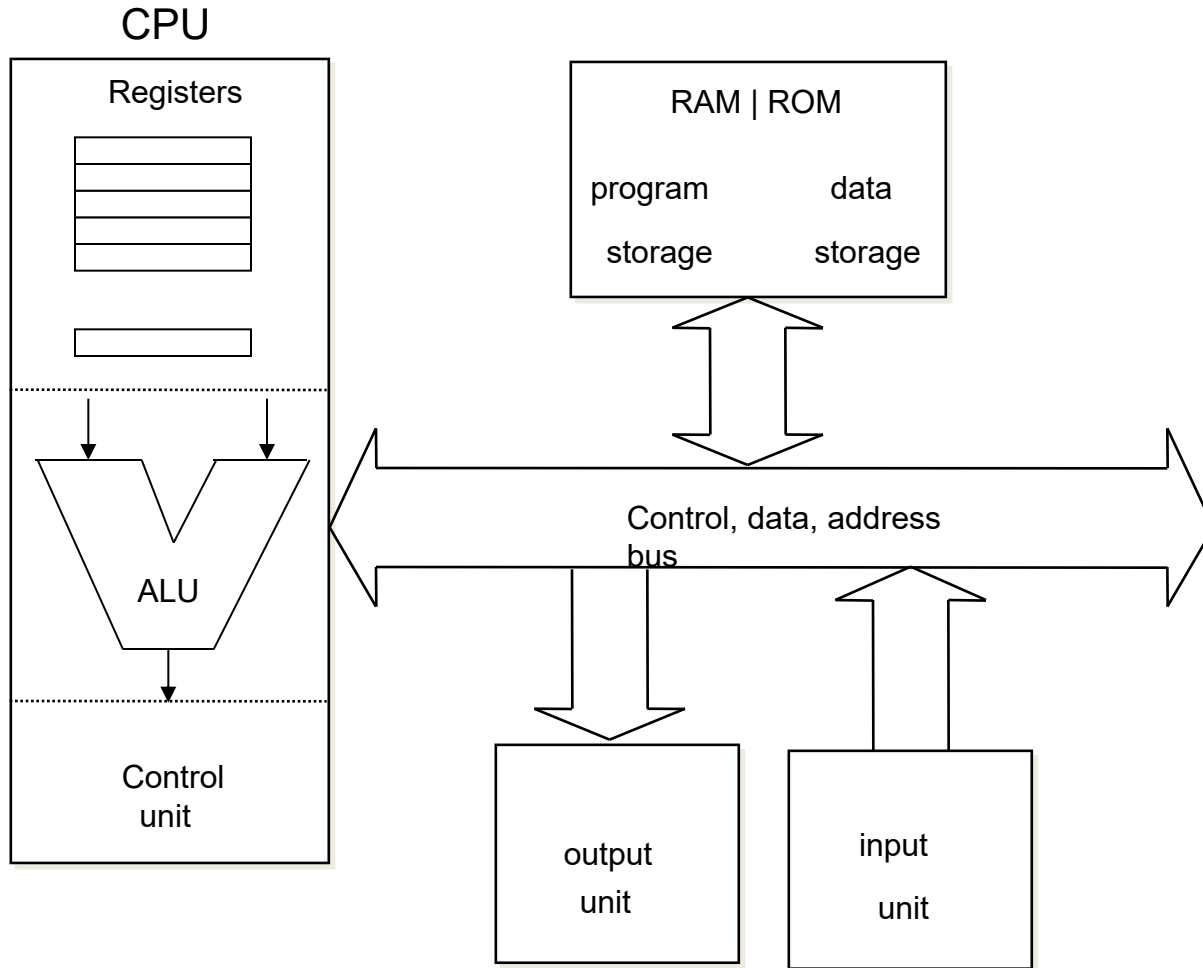
High-level languages

- Fortran
- Basic
- Visual Basic
- C
- C++
- Java

Advantages of Assembly languages

- Efficient
 - It is so close to machine language.
 - Produces a faster, shorter machine language program.
- Some operations, such as reading or writing to specific memory locations and I/O ports can be done easily in assembly language but may be impossible at a higher level.
- It also provides the way to realize how the computer “thinks” and performs the inner operations.

Computer Hardware Organization





Registers

- Information inside the microprocessor is stored in registers.
- Even though the processor can operate on data stored in memory, the same instruction is faster if the data are stored in registers.

Types

- Registers are classified into three categories according to the functions they perform. Each register is 16 bit.
 - Data registers
 - Hold data for an operation
 - Address registers
 - Hold the address of an instruction and data
 - Status registers
 - Keep the current status of the processor

Data Registers

- 8086 has four data registers for general data manipulation. They are:
 - AX (Accumulator Register)
 - BX (Base Register)
 - CX (Count Register)
 - DX (Data Register)
- The high and low bytes of the data registers can be accessed separately. The high byte of AX is called AH and the low byte is AL.

Data Registers

- **AX (Accumulator Register....AH+AL)**
 - Use in arithmetic, logic and data transfer instructions because its use generates the shortest machine code.
 - In multiplication and division operations, one of the numbers involved must be in AX or AL.
- **BX (Base Register.....BH+BL)**
 - Serves as an address register

Data Registers

- CX (Count Register.....CH+CL)
 - Use in program loop constructions....serves as loop counter
 - Use in REP (repeat), string operations, shift and rotate operations.
- DX (Data Register.....DH+DL)
 - Use in multiplication and division
 - Use in I/O operations

Definition of the Microprocessor

The microprocessor is a programmable device that takes in numbers, performs on them arithmetic or logical operations according to the program stored in memory and then produces other numbers as a result.

Definition (Contd.)

- Lets expand each of the underlined words:
 - **Programmable device**: The microprocessor can perform different sets of operations on the data it receives depending on the sequence of instructions supplied in the given program.
By changing the program, the microprocessor manipulates the data in different ways.
 - **Instructions**: Each microprocessor is designed to execute a specific group of operations. This group of operations is called an instruction set. This instruction set defines what the microprocessor can and cannot do.

Definition (Contd.)

- **Takes in:** The data that the microprocessor manipulates must come from somewhere.
 - It comes from what is called “input devices”.
 - These are devices that bring data into the system from the outside world.
 - These represent devices such as a keyboard, a mouse, switches, and the like.

Definition (Contd.)

- **Numbers:** The microprocessor has a very narrow view on life. It only understands binary numbers.

A binary digit is called a bit (which comes from binary digit).

The microprocessor recognizes and processes a group of bits together. This group of bits is called a “word”.

The number of bits in a Microprocessor’s word, is a measure of its “abilities”.

Definition (Contd.)

– Words, Bytes, etc.

- The earliest microprocessor (the Intel 8088 and Motorola's 6800) recognized 8-bit words.
 - They processed information 8-bits at a time. That's why they are called "8-bit processors". They can handle large numbers, but in order to process these numbers, they broke them into 8-bit pieces and processed each group of 8-bits separately.
- Later microprocessors (8086 and 68000) were designed with 16-bit words.
 - A group of 8-bits were referred to as a "half-word" or "byte".
 - A group of 4 bits is called a "nibble".
 - Also, 32 bit groups were given the name "long word".
- Today, all processors manipulate at least 32 bits at a time and there exists microprocessors that can process 64, 80, 128 bits

Definition (Contd.)

– Arithmetic and Logic Operations:

- Every microprocessor has arithmetic operations such as add and subtract as part of its instruction set.
 - Most microprocessors will have operations such as multiply and divide.
 - Some of the newer ones will have complex operations such as square root.
- In addition, microprocessors have logic operations as well. Such as AND, OR, XOR, shift left, shift right, etc.
- Again, the number and types of operations define the microprocessor's instruction set and depends on the specific microprocessor.

Definition (Contd.)

- Stored in memory :
 - First, what is memory?
 - Memory is the location where information is kept while not in current use.
 - Memory is a collection of storage devices. Usually, each storage device holds one bit. Also, in most kinds of memory, these storage devices are grouped into groups of 8. These 8 storage locations can only be accessed together. So, one can only read or write in terms of bytes to and from memory.
 - Memory is usually measured by the number of bytes it can hold. It is measured in Kilos, Megas and lately Gigas. A Kilo in computer language is $2^{10}=1024$. So, a KB (KiloByte) is 1024 bytes. Mega is 1024 Kilos and Giga is 1024 Mega.

Definition (Contd.)

– Stored in memory:

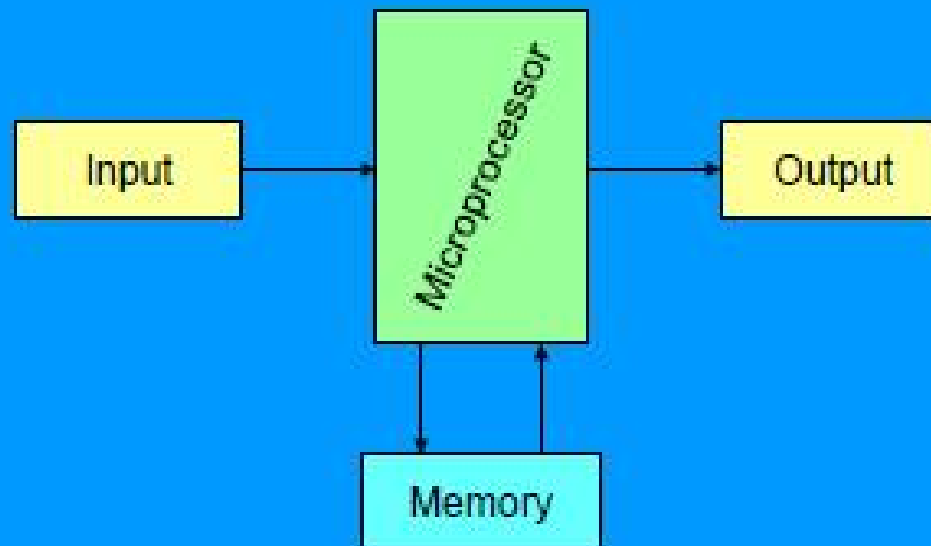
- When a program is entered into a computer, it is stored in memory. Then as the microprocessor starts to execute the instructions, it brings the instructions from memory one at a time.
- Memory is also used to hold the data.
 - The microprocessor reads (brings in) the data from memory when it needs it and writes (stores) the results into memory when it is done.

Definition (Contd.)

- **Produces:** For the user to see the result of the execution of the program, the results must be presented in a human readable form.
 - The results must be presented on an output device.
 - This can be the monitor, a paper from the printer, a simple LED or many other forms.

A Microprocessor-based system

From the above description, we can draw the following block diagram to represent a microprocessor-based system:

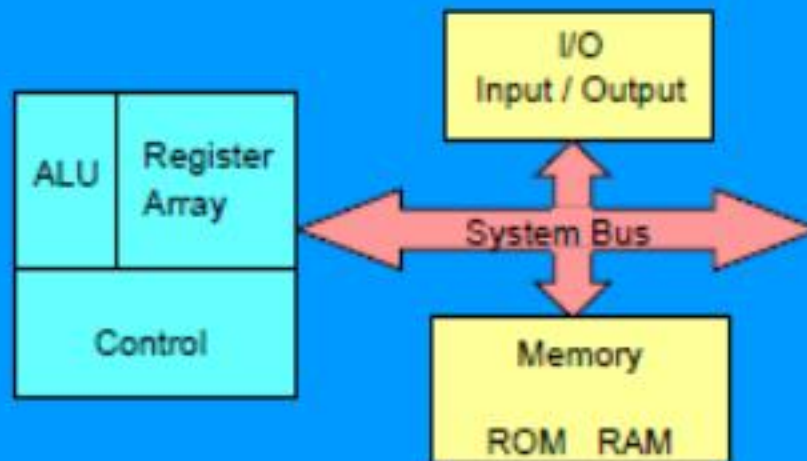


Inside The Microprocessor

- Internally, the microprocessor is made up of 3 main units.
 - The Arithmetic/Logic Unit (ALU)
 - The Control Unit.
 - An array of registers for holding data while it is being manipulated.

Organization of a microprocessor-based system

- Let's expand the picture a bit.



Memory

- Memory stores information such as instructions and data in binary format (0 and 1). It provides this information to the microprocessor whenever it is needed.
- Usually, there is a memory “sub-system” in a microprocessor-based system. This sub-system includes:
 - The registers inside the microprocessor
 - Read Only Memory (ROM)
 - used to store information that does not change.
 - Random Access Memory (RAM) (also known as Read/Write Memory).
 - used to store information supplied by the user. Such as programs and data.

Memory

- To execute a program:
 - the user enters its instructions in binary format into the memory.
 - The microprocessor then reads these instructions and whatever data is needed from memory, executes the instructions and places the results either in memory or produces it on an output device.

The three cycle instruction execution model

- To execute a program, the microprocessor “reads” each instruction from memory, “interprets” it, then “executes” it.
- To use the right names for the cycles:
 - The microprocessor fetches each instruction,
 - decodes it,
 - Then executes it.
- This sequence is continued until all instructions are performed.

Basic element of a computer are

- CPU
- Memory
- Input/Output (I/O) units.

Translates

instructions

Perform

arithmetic of logic operations

Temporarily stores

instructions & data in its

internal high-speed registers

CPU

Memory : Stores Program & Data

I/O Unit : Interfaces the computer with external device such as Keyboard and Display.

With the advent of semiconductor technology, it is possible to integrate CPU in a chip.

Result : Microprocessor

Technology : MOS (both by Intel & Motorola)

Single chip microcomputers are also available in which microprocessors, memory and I/O are all fabricated the same chip. (limited capabilities ideal for peripheral controllers)

Single Chip microcomputers —> Microcontrollers

Microcontrollers —> Automotive systems,
home entertainment system.

Evolution of the Microprocessor

- Intel corporation introduced 1st **Microprocessor** 4004 (4bit) in 1971.
- The 4004 evolved from a development effort while designing a Calculator Chip set.
- Soon after 4004 appeared Commercially, 3 other microprocessors use introduced

- 
- Rockwell International Intel 8 bit 8008 & National Semiconductor 16 bit Imp-16.
 - Microprocessor between 1971 and 1973 :
1st Generation system Designed with PMOS technology.
 - This technology provides
 - Low cost
 - Slow Speed
 - Low input current &
 - Was not compatible with TTL

- After 1973, 2nd Generation Microprocessor such as,
MOS technology 6502
Motorola 6800 & 6809
Intel 8085 & Zilog Z80 evolved
- There 8 bit processors were fabricated using NMOS technology.
- NMOS process offers faster speed and higher density than PMOS & is TTL-Compatible.
- After 1978, the 3rd Generation microprocessors were introduced.
- These processors are 16 bits wide (16 bit ALU) & include: Intel 8086/80186/80286 &
Motorola 68000/68010
Technology : HMOS (High Density MOS)



- HMOS provides the following advantages over NMOS

- ❖ Speed-Power-Product (SPP) is 4 times better than NMOS

- ❖Circuit densities provided by HMOS are approximately twice than of NMOS.

- Later, Intel utilized the HMOS technology to fabricate 8085A. High Speed Version of 8085 A is 8085 AH.



- In 1980, 4th generation microprocessors are evolved.

- ❖ Intel introduced the first commercial 32 bit microprocessor the problematic Intel 432. This Processor was eventually discontinued by Intel.

- ❖ Since 1985, more 32 bit microprocessors have been introduced. These include:

Motorola's MC 68020/68030/68040

Intel's 80386/80486

Intel Pentium processors.



- These processors are fabricated using Low-power version of the HMOS technology called the HCMOS.

- The performance offered by the 32 bit processor is make comparable to that of superminicomputers such as DEC's VAX 11/750 & VAX 11/780.

- Both Intel & Motorola introduced 32 bit RISC processor (Intel 80960 & Motorola MC88100/PowerPC) with simplified instruction sets.

- **Purpose of RISC processors** : Maximize speed by reducing clock cycles per instruction.
- RISC microprocessor have hardwired instructions sets like non RISC microprocessor (Such as Intel 8085 & MOS 6502).
- This means that for every instruction, there exist actual, Physical connections that provide the desired instruction decoding.
- The microprocessor, therefore does not use valuable clock cycle (machine cycle) in instruction decoding.

- 
- Since RISC type microprocessor are hardwired, instructions may be executed simultaneously (as long as instruction do not share the same register).
 - This technique is pipelining.
 - The processor continues to fetch instructions even though it has not executed the present instructions.
 - At this stage, the instructions are intelligently fed into the instructing execution unit.
 - Registers are checked whether second instruction can be executed.

- 
- The RISC microprocessor can execute as many as 5 instructions simultaneously.
 - The trend in microprocessor is implementation of more on-chip functions and for improvement of the speeds of memory and I/O devices.

The work that each instruction of the RISC machine performs is simple and straight forward. Thus, the time required to execute each instruction can be shortened and the number of cycles reduced. Typically the instruction execution time is divided in five stages, *machine cycles*, and as soon as processing of one stage is finished, the machine proceeds with executing the second stage. However, when the *stage* becomes free it is used to execute the same operation that belongs to the next instruction. The operation of the instructions is performed in a *pipeline* fashion, similar to the assembly line in the factory process. Typically those five *pipeline stages* are:



IF – Instruction Fetch

ID – Instruction Decode

EX – Execute

MA – Memory Access

WB – Write Back

By overlapping the execution of several instructions in a pipeline fashion (as shown in Fig. 1.), RISC achieves its inherent execution parallelism which is responsible for the performance advantage over the Complex Instruction Set Architectures (CISC).

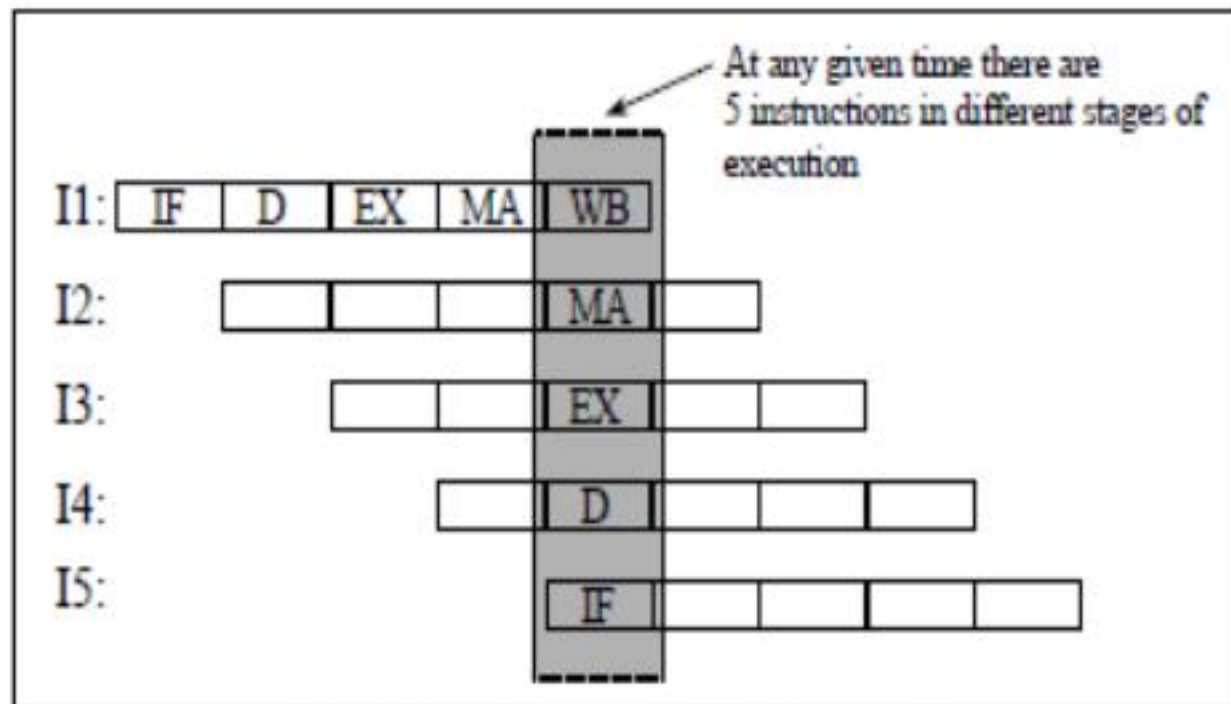


Fig. 1. Typical five stage RISC pipeline

Pipelining

Overview

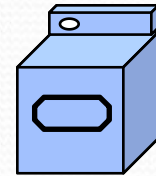
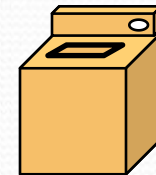
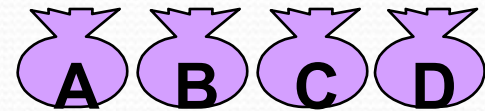
- Pipelining is widely used in modern processors.
- Pipelining improves system performance in terms of throughput.
- Pipelined organization requires sophisticated compilation techniques.

Making the Execution of Programs Faster

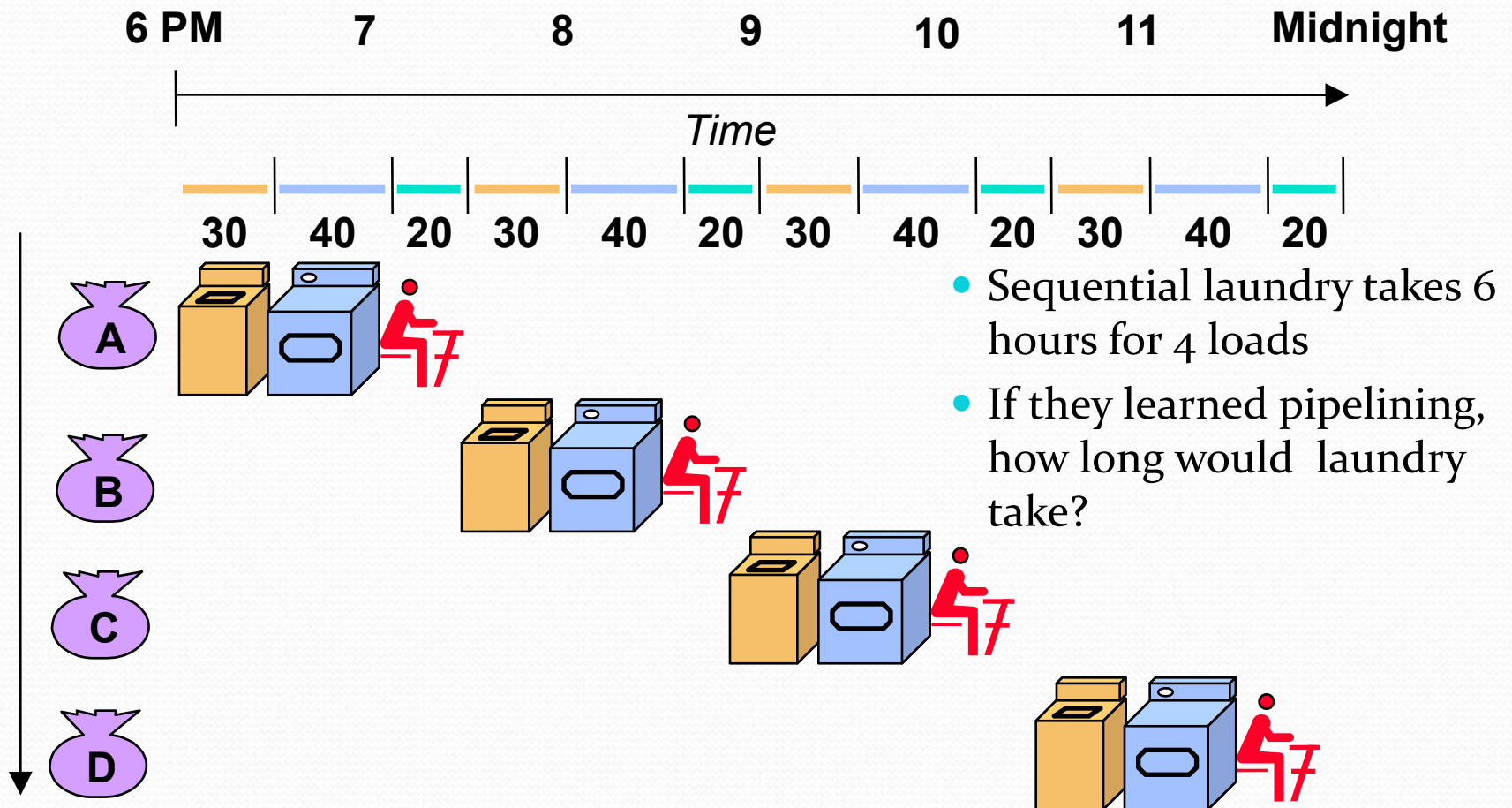
- Use faster circuit technology to build the processor and the main memory.
- Arrange the hardware so that more than one operation can be performed at the same time.
- In the latter way, the number of operations performed per second is increased .

Traditional Pipeline Concept

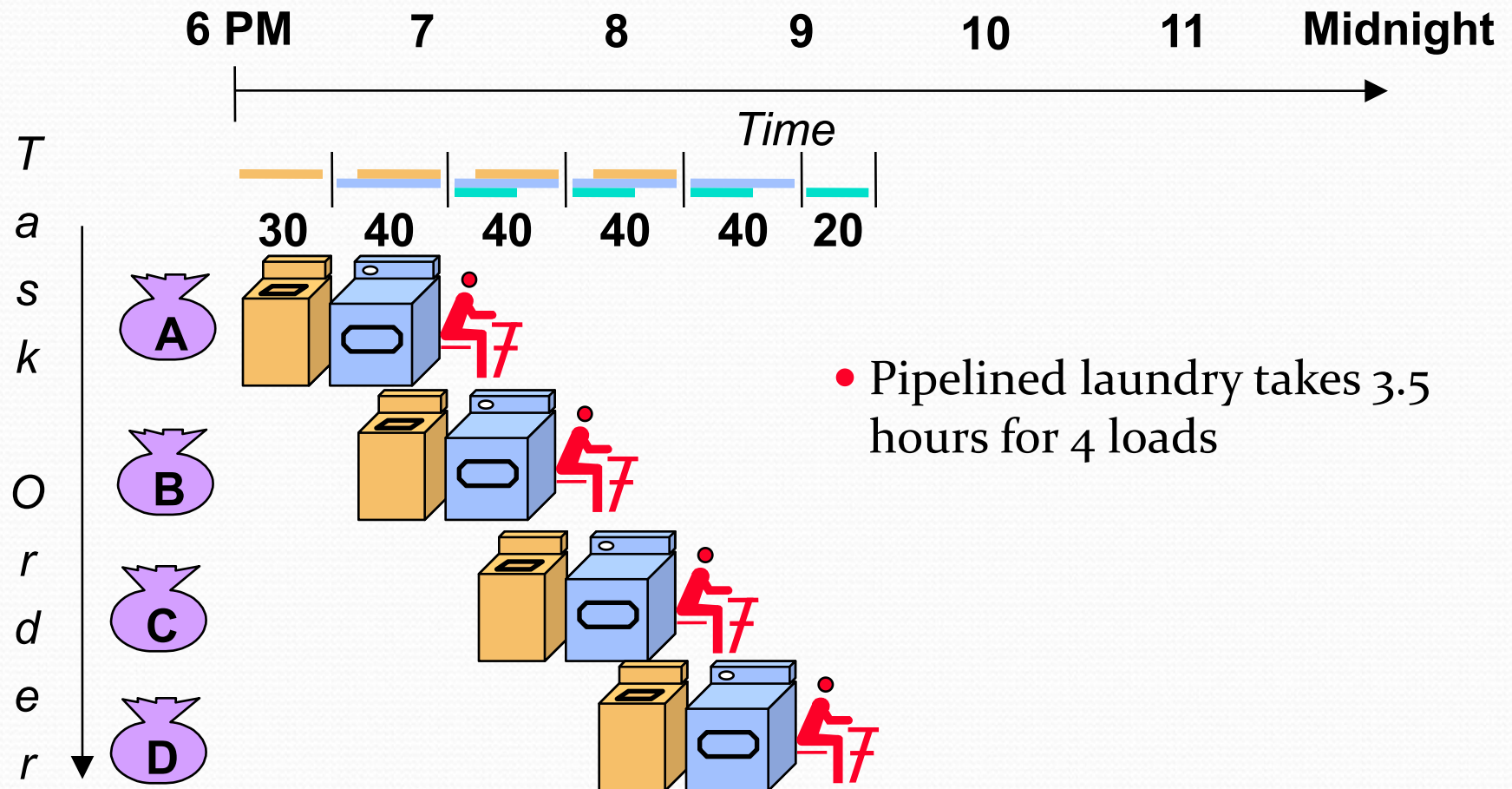
- Laundry Example
- Ann, Brian, Cathy, Dave each have one load of clothes to wash, dry, and fold
- Washer takes 30 minutes
- Dryer takes 40 minutes
- “Folder” takes 20 minutes



Traditional Pipeline Concept



Traditional Pipeline Concept



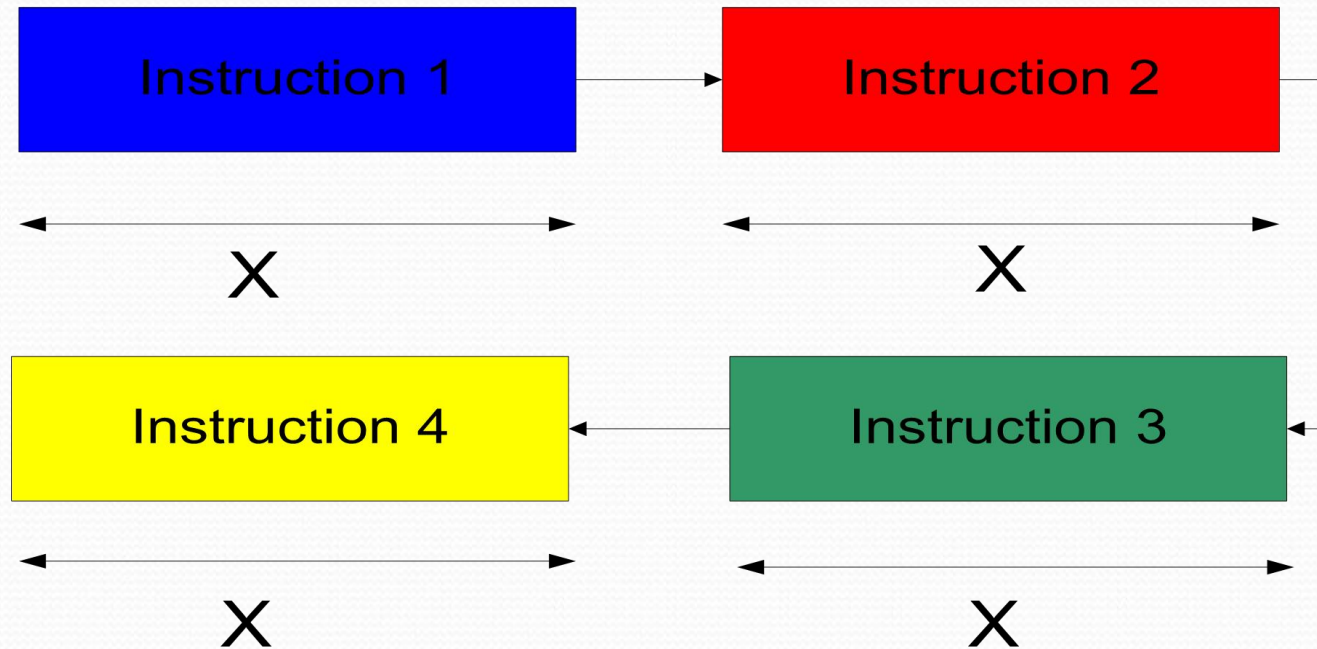
What is Pipelining

- A technique used in advanced microprocessors where the microprocessor begins executing a second instruction before the first has been completed.
- A Pipeline is a series of stages, where some work is done at each stage. The work is not finished until it has passed through all stages.
- With pipelining, the computer architecture allows the next instructions to be fetched while the processor is performing arithmetic operations, holding them in a buffer close to the processor until each instruction operation can be performed.

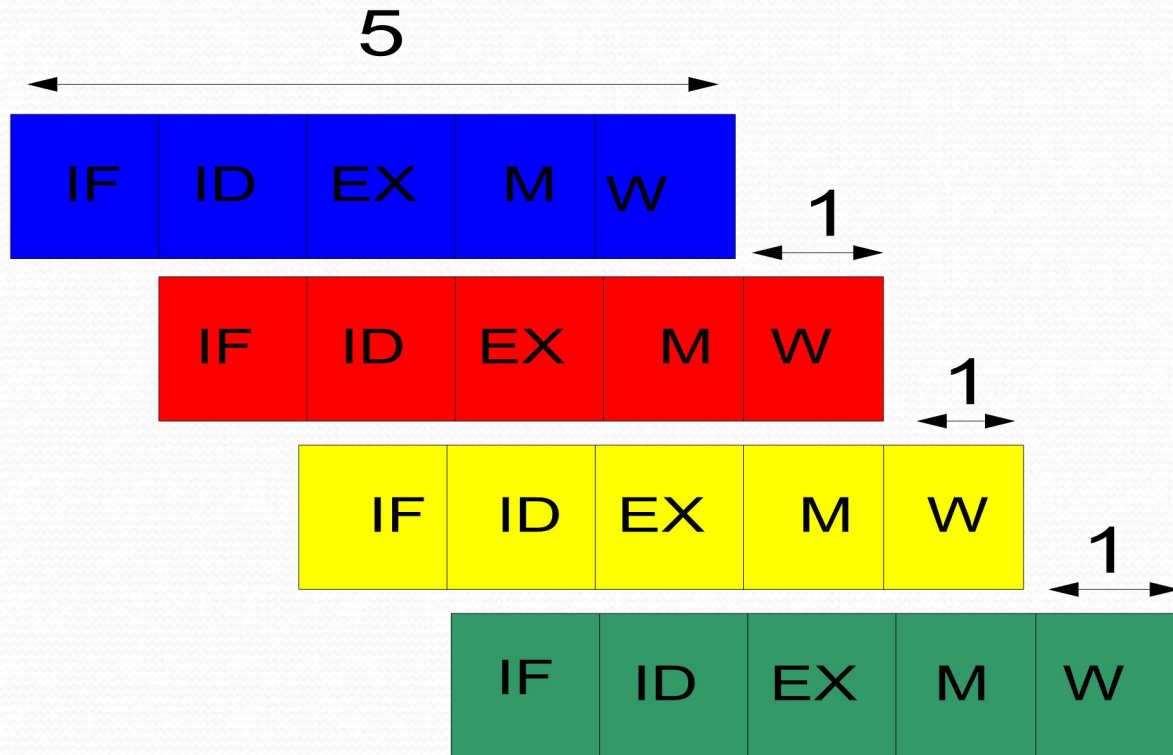
How Pipelines Works

- The pipeline is divided into segments and each segment can execute its operation concurrently with the other segments. Once a segment completes an operation, it passes the result to the next segment in the pipeline and fetches the next operations from the preceding segment.

Example



Four sample instructions, executed linearly



Four Pipelined Instructions

Instructions Fetch

- The instruction Fetch (IF) stage is responsible for obtaining the requested instruction from memory. The instruction and the program counter (which is incremented to the next instruction) are stored in the IF/ID pipeline register as temporary storage so that may be used in the next stage at the start of the next clock cycle.

Instruction Decode

- The Instruction Decode (ID) stage is responsible for decoding the instruction and sending out the various control lines to the other parts of the processor. The instruction is sent to the control unit where it is decoded and the registers are fetched from the register file.

Execution

- The Execution (EX) stage is where any calculations are performed. The main component in this stage is the ALU. The ALU is made up of arithmetic, logic and capabilities.

Memory and IO

- The Memory and IO (MEM) stage is responsible for storing and loading values to and from memory. It also responsible for input or output from the processor

Write Back

- The Write Back (WB) stage is responsible for writing the result of a calculation, memory access or input into the register file.